

Assignment 1: Data Pre-processing

Assignment Instructions

Extend your **Lab 2 notebook** into a concise technical report (maximum **four pages of written content**, excluding the notebook). The report should be written for the **Head of Credit Risk** at the lending institution, focusing on the quality, reliability, and reproducibility of your machine learning solution.

Your report should include the following sections:

1. Data Quality Audit (5 Marks)

Conduct a comprehensive audit of the dataset and identify the **ten columns with the most significant data quality issues** (refer to **Figure 2.1**).

For each column, provide:

- The data quality issue identified.
- The evidence supporting your conclusion (e.g., summary statistics, missing value analysis, duplicate counts, diagnostic plots, or validation checks).
- The missing-data classification where applicable (**MCAR, MAR, or MNAR**).
- The remediation strategy you selected and why it is appropriate.

Present your findings in a well-organised table.

2. Feature Dictionary (5 Marks)

Create and document **at least 15 engineered features** drawn from a **minimum of four feature families** listed in **Table 2.2**.

For each engineered feature, include:

- **Feature name**
- **Definition** (what the feature represents)
- **Computation window** (the time period over which it is calculated)
- **Upstream fields (data lineage)** used to construct the feature
- **Write time** of every upstream field (when each source field becomes available)

- A brief **fairness note**, where relevant (see **Remark 2.3**), describing any potential bias or ethical consideration associated with the feature.
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3. Leakage Assessment Memo (6 Marks)

Prepare a short technical memo describing the **two intentionally planted sources of data leakage**.

For each leakage source:

- Identify where it occurs.
 - Explain **how** the leakage happens by referring to the **six causes of data leakage discussed in Section 2.7.2**.
 - Quantify the impact by comparing the model performance before and after the leakage is removed (for example, the change in offline **ROC-AUC**).
 - Recommend a practical process or governance change that would prevent this type of leakage in future model development.
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4. Reproducible Machine Learning Pipeline (4 Marks)

Develop a **single, end-to-end Scikit-learn Pipeline** that performs all preprocessing and model training.

Your submission must:

- Perform **all preprocessing steps within the pipeline**.
 - Evaluate the model using **group-aware cross-validation**.
 - Report the final cross-validated **ROC-AUC**.
 - Save the trained pipeline using **joblib**.
 - Ensure the saved pipeline can be reloaded and executed successfully by the marker without modification.
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Important Requirements

- Every conclusion or design decision must be supported with **evidence**, not assumptions.

- Any preprocessing performed **outside the cross-validation pipeline** will be treated as **data leakage** and will result in the loss of the pipeline marks, regardless of the model's performance.
- The emphasis of this assignment is on producing a solution that is **correct, reproducible, well-documented, and suitable for deployment**, rather than simply achieving the highest predictive accuracy.